FRANKFURT UNIVERSITY OF APPLIED SCIENCES

ML 24/25 IMPLEMENT THE VISUALIZATION OF PERMANENCE VALUE

SUBMITTED BY:

MD MUSHFIQUL ISLAM
1438422

PROLOY SEN
1517624

REZOYANA ISLAM BONNA
1502329

MASTER OF ENGINEERING
INFORMATION TECHNOLOGY

OVERVIEW

1. Objective
2. INTRODUCTION
3. Methodology
4. Reconstruction Method in NeocortexAPI
5. Graphical representation of Draw Heatmap Function
6. Visualization Technique
7. Visualization Result
8. Validation of Result
9. Similarity for Image Inputs
10. Conclusion

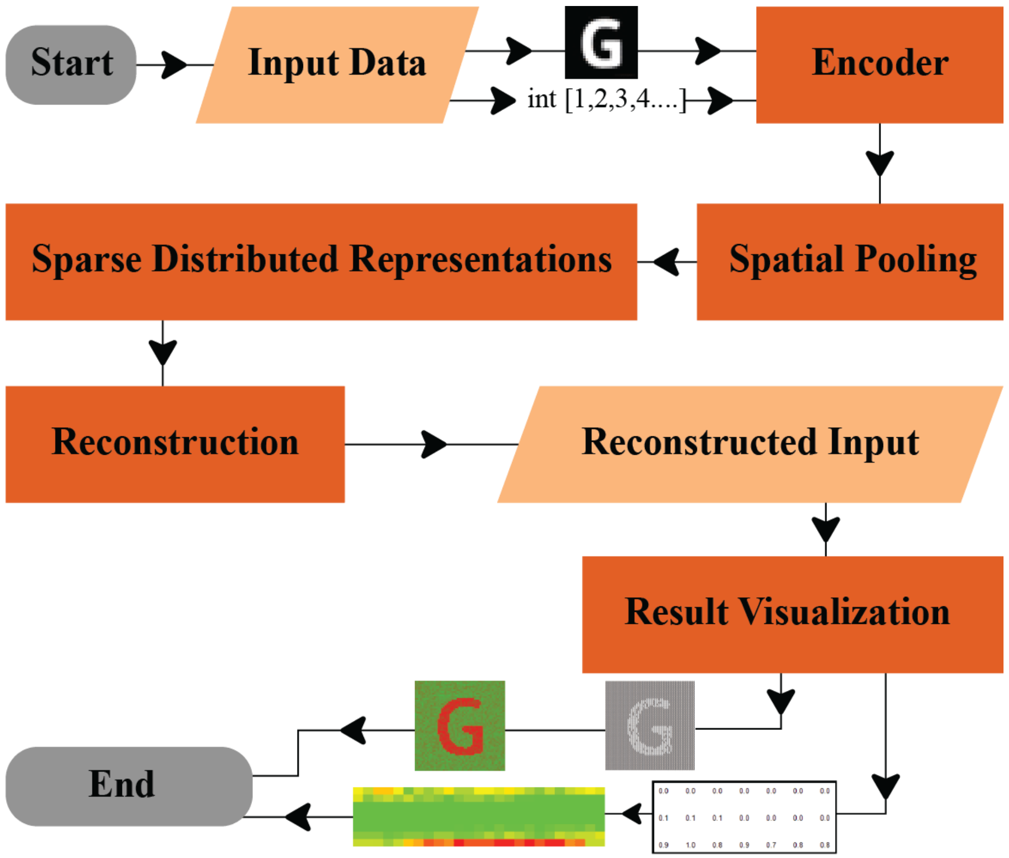
OBJECTIVE

- Develop an improved visualization approach for tracking permanence value dynamics in the HTM Spatial Pooler.
- Enhance interpretability and optimization of HTM models.
- Use Neocortex API's Reconstruct method to track permanence changes over time.
- Apply advanced visualization techniques to represent permanence distributions.
- Extend the approach to image data for evaluating its role in visual pattern recognition & reconstruction.
- Provide better insights into learning dynamics and synaptic stability in HTM networks.

INTRODUCTION

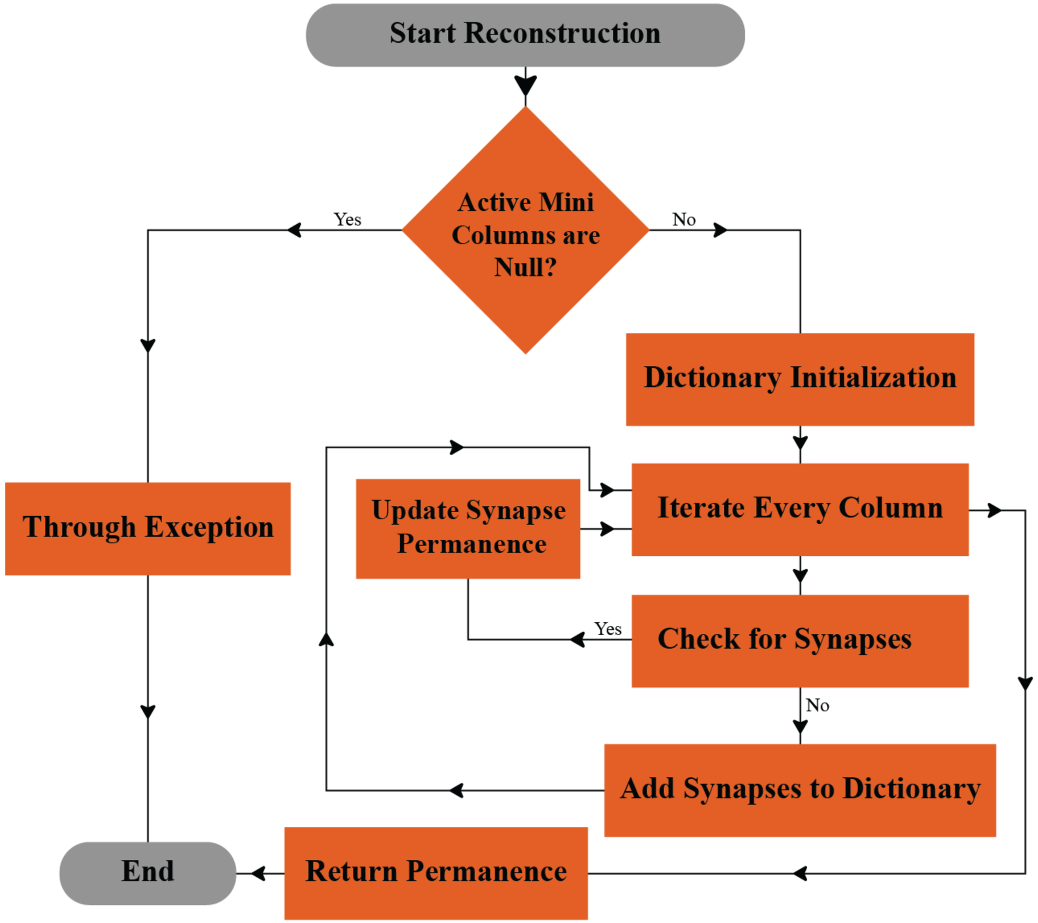
- Hierarchical Temporal Memory (HTM) is inspired by the human neocortex and specializes in recognizing temporal patterns using Sparse Distributed Representations (SDRs).
- The Spatial Pooler (SP) converts raw input data into SDRs, mimicking the brain's encoding mechanism for efficient pattern recognition & storage.
- Our research focuses on the Spatial Pooler and Neocortex API's Reconstruct() function, analyzing their role in accurate input reconstruction and permanence evolution visualization.
- Permanence values determine synaptic stability and evolve with learning—visualizing them enhances our understanding of the HTM model
- We explore two permanence representations: 1.Numerical inputs 2. Image data inputs
- These visualizations provide deeper insights into learning dynamics and encoded representations in HTM.

METHODOLOGY



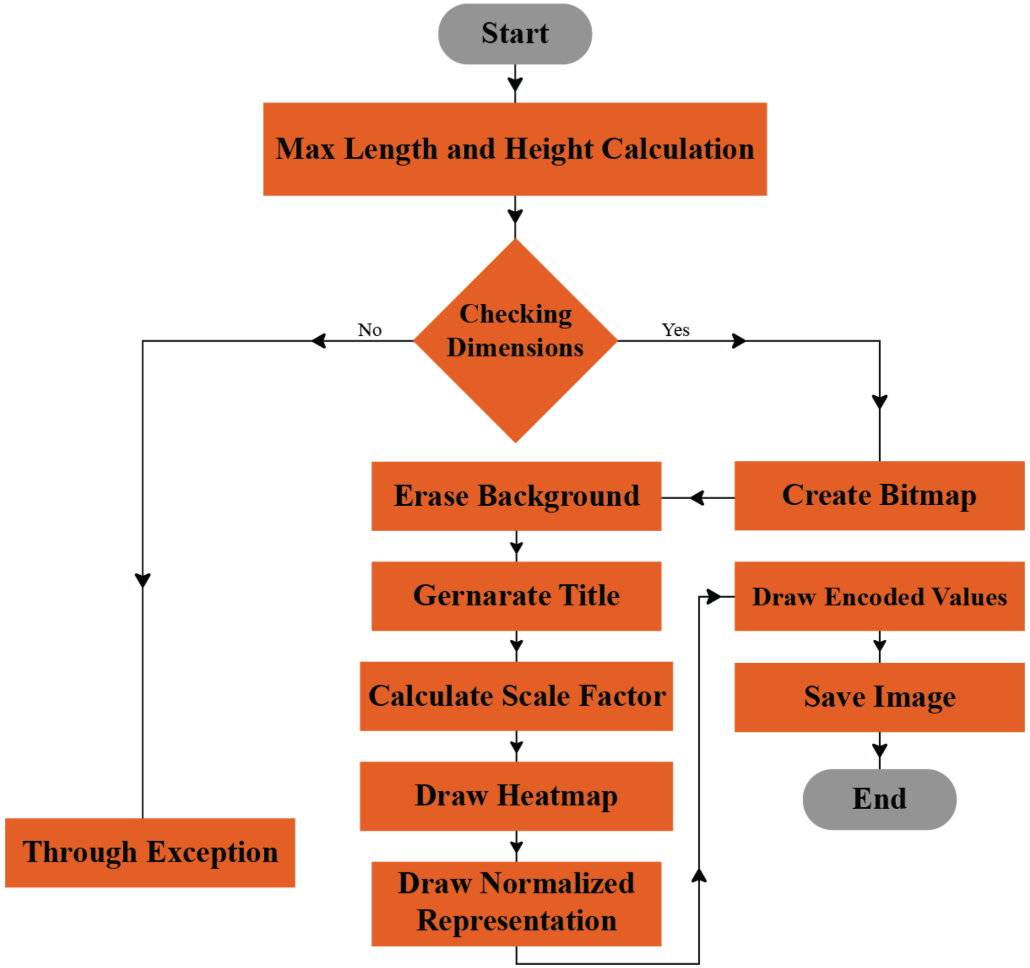
- **Input Data** → Encoded into a binarized form.
- **Spatial Pooling** → Creates SDRs by detecting patterns.
- **Reconstruction** → Converts SDRs back to approximate input.
- **Result Visualization** → Displays the reconstructed data.

RECONSTRUCTION METHOD IN NEOCORTEX API



1. **Start Reconstruction** → Check if active mini-columns are null.
   - **If Yes** → Throw exception and end.
   - **If No** → Initialize dictionary and iterate columns.
2. **Check for Synapses**:
   - **Found** → Update permanence.
   - **Not found** → Add to dictionary.
3. **Update Synapse Permanence** → Adjust and loop if needed.
4. **Return Permanence** → End process.

DRAW HEATMAP FUNCTION



- **Start:** Process begins.
- **Max Length & Height Calculation:** Determines size constraints.
- **Checking Dimensions (Decision Point):**
  - **Yes** → Create Bitmap → Erase Background.
  - **No** → Trigger exception handling.
    - **Exception Handling:** If dimensions are invalid, exit process.
- **Generate Title** → Add labels/titles.
- **Calculate Scale Factor** → Adjust size.
- **Draw Heatmap** → Visualize data.
- **Draw Normalized Representation** → Final processing.
- **Draw Encoded Values** → Add encoded data.
- **Save Image** → Store final output.
- **End.**

VISUALIZATION TECHNIQUE

- Encoded Input Comparison
  - Compares the original input with the reconstructed input.
  - Assesses how accurately the HTM model retains key information.

- Similarity Analysis
  - Measures the correlation between input and reconstructed output.
  - Evaluates pattern recognition efficiency and learning accuracy.

VISUALIZATION TECHNIQUES

- Normalization with Threshold
  - Adjusts permanence values to a defined threshold for better interpretability.
  - Highlights significant changes in synaptic permanence over time.

- Heatmap Representation
  - Visualizes permanence values across different data points.
  - Helps identify patterns and synaptic stability in the Spatial Pooler.

COMPARING ORIGINAL INPUT AND RECONSTRUCTED MATRICS

Encoded Input Matrics

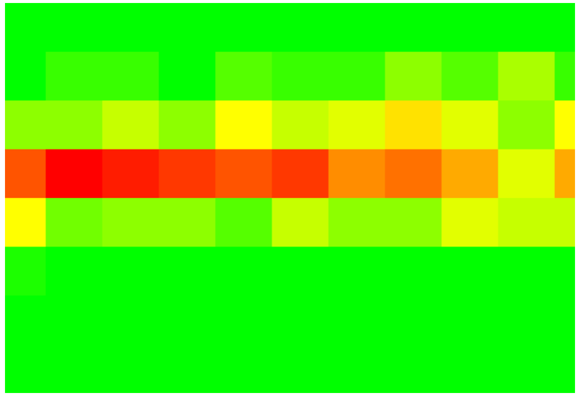
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	1	1	0	0	0	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	0	1	0	1	1	0	0
0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reconstruced Input Matrics

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

RESULT VISUALIZATION

For Numerical Input



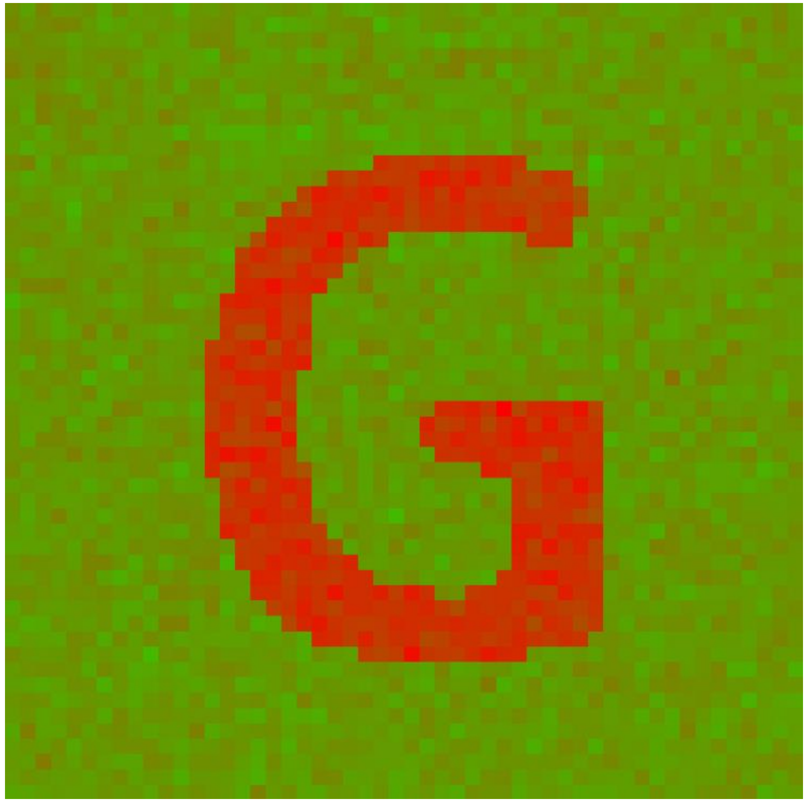
Heatmap

0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.1	0.0	0.2	0.1	0.1	0.3	0.2	0.3	0.1	0.2	0.3	0.2	0.1	0.3	0.3	0.2	0.2
0.3	0.4	0.2	0.2	0.2	0.3	0.3	0.3	0.4	0.3	0.5	0.4	0.4	0.6	0.4	0.3	0.5	0.5	0.3	0.4	0.3	0.5	0.7	0.5	0.9
0.7	0.8	0.7	1.0	0.9	0.8	0.8	1.0	0.9	0.9	0.8	0.9	0.7	0.8	0.7	0.4	0.7	0.7	0.4	0.7	0.3	0.6	0.5	0.2	0.2
0.3	0.3	0.4	0.4	0.4	0.3	0.5	0.2	0.3	0.3	0.2	0.4	0.3	0.3	0.4	0.4	0.4	0.2	0.1	0.1	0.1	0.1	0.2	0.1	0.2
0.1	0.1	0.1	0.1	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

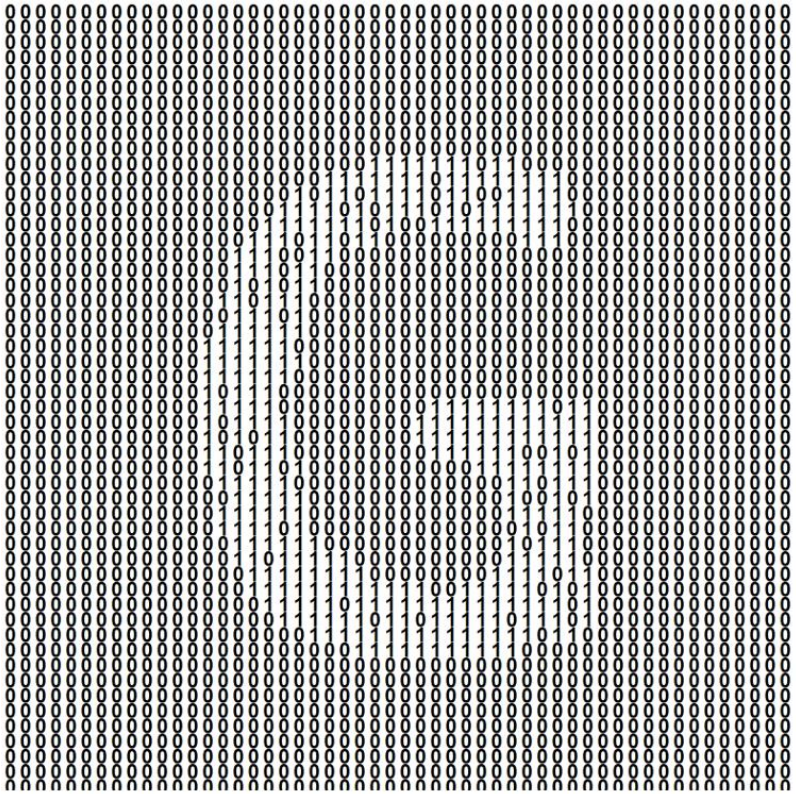
Matrics

RESULT VISUALIZATION

For Image Input

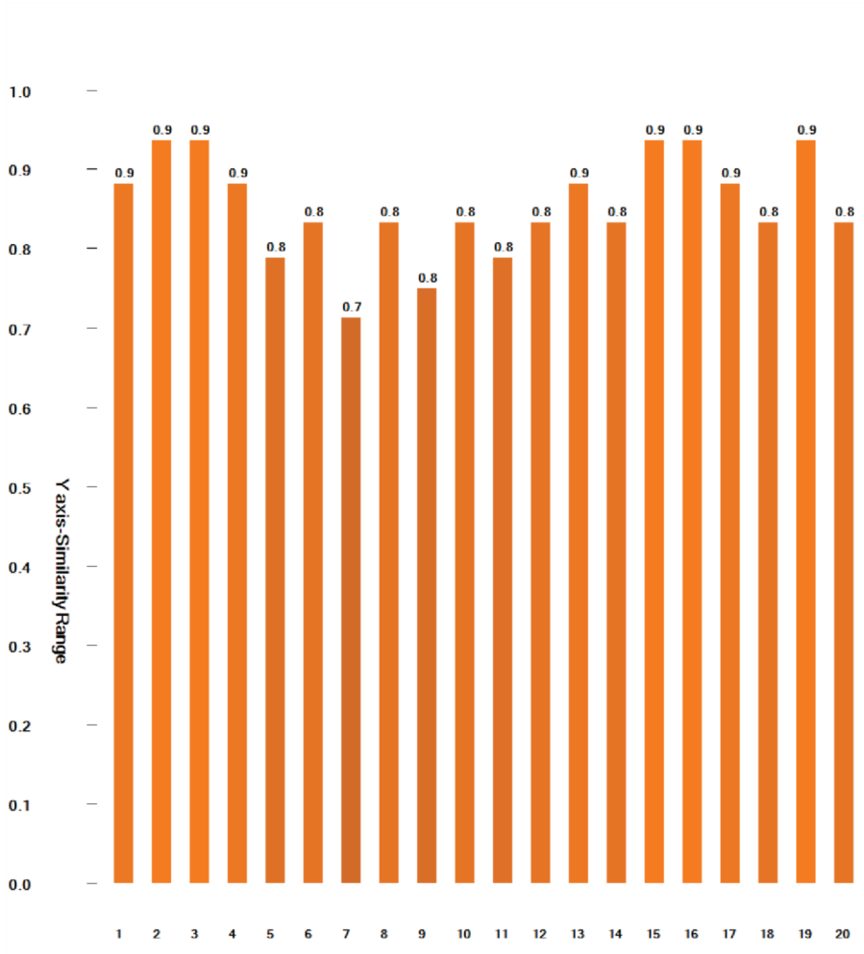


Heatmap

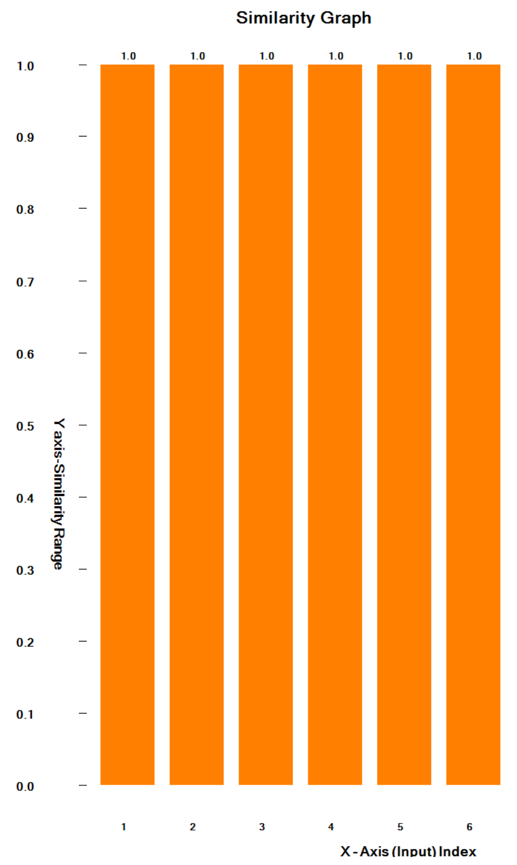


Matrics

SIMILARITY GRAPH



The similarity graphs of numerical Inputs



Similarity Graph for the Image Input

CONCLUSION

- Visualization and Analysis
  - Successfully implemented HTM-based reconstruction and visualization.
  - Used heatmaps and matrices to analyze permanence value distribution.
  - Similarity graphs provided quantitative insights into reconstruction accuracy.
- Impact and Future Scope
  - Effective for both numerical and image-based inputs.
  - Enhances interpretability of HTM learning dynamics.
  - Establishes a foundation for future improvements in HTM-based models.

THANK YOU